Fiscal Year:	FY 2006	**Task Last Updated:**	FY 06/22/2006
PI Name:	Ronca, April Elizabeth Ph.D.		
Project Title:	Hypergravity Effects On The Maternal-Fetal System		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology (2) Developmental Biology (3) Musculoskeletal Biology (4) Neurobiology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
PI Email:	april.e.ronca-1@nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	650.400.6019
Organization Name:	NASA Ames Research Center		
PI Address 1:	Space Biosciences Research Branch		
PI Address 2:	MS 261		
PI Web Page:			
City:	Moffett Field	**State:**	CA
Zip Code:	94035	**Congressional District:**	18
Comments:	November 2019: PI is located at NASA Ames Research Center and remains affiliated with Wake Forest University School of Medicine		
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Space Biology 01-OBPR-06
Start Date:	12/01/2002	**End Date:**	08/31/2008
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	5	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 8/31/2008 per A. Chu (8/07) 8/06--funding reinstated through 8/31/2007, per J. Cohen (HQ). Retaining original start date of 12/01/2002.		
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Wade, Charles Ph.D. (US Army Institute of Surgical Research)		
Grant/Contract No.:	NNA04CK83A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This project will advance our understanding of gravity's influence(s) on mammalian pregnancy, parturition and offspring development and their inter-relations. Through a series of integrative experiments, we seek to uncover similarities between prenatal exposure to increased (or hyper-) gravity and other intrauterine events, such as prenatal nutrients and oxygen that 'program', or set the stage for developmental outcomes and adult disease. The fetal programming hypothesis states that prenatal events set into motion a series of physiological alterations in the developmental program that may translate into adult disorders, in particular, cardiovascular disease, diabetes and mental illness. In the current project, Norway rats will be exposed to 2-g or 1.5-g hypergravity to establish continuous centrifugation as a model for fetal programming of adult disease. Non-pregnant females will be adapted to hypergravity prior to mating to eliminate effects of the adaptation process on fetal offspring. We have previously established reduced birth weights of rats exposed to 2-g or 1.5-g centrifugation during prenatal development. We have also observed decreased numbers of labor contractions in dams that underwent pregnancy during 2-g or 1.5-g centrifugation. Behavioral, physiological and molecular methods will be combined to ascertain mechanisms underlying effects of centrifugation on the maternal-fetal system. We will characterize and compare effects of hypergravity on uterine contractile activity during gestation and birth using intrauterine pressure measurement, hormonal and behavioral measures. Studies of fetal offspring at the time of birth will control exposure to uterine contractions by either administering specific numbers of artificially-produced contractions or by completely eliminating exposure to them. Comparisons will be made between offspring that undergo gestation in hypergravity and those exposed to bilateral occlusion of the uterine arteries during late pregnancy, the latter being an established model of fetal programming. Labor contractions will be related to neonatal variables, namely postpartum breathing and oxygenation, suckling and metabolic activity. Molecular analyses will focus on the expression of specific genes that are reactive to sensory simulation, the early, immediate genes, c-fos and c-jun in brain and adrenal of the newborn rats associated with birth. Longitudinal studies of the offspring will focus on adult cardiovascular responses to stress.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The newly emerging field of fetal programming, the idea that perinatal events direct or 'program' adult health and disease states, is uncovering startling insights into relationships between intrauterine events and adult health problems, including cardiovascular disease, diabetes and mental health disorders. Numerous epidemiological and animal studies provide unequivocal evidence for a strong association between low birth weight and adult cardiovascular disease. The goal of the proposed research is to determine whether hypergravity exposure of developing offspring is an appropriate model for fetal programming and can be used to further scientific understanding of fetal programming of adult disease.
Task Progress:	In recent years, the number of female astronauts has grown immensely. As we move forward with the Vision for Space Exploration, women are expected to increase their involvement in the Space Program and particularly, long-duration missions. The new space initiative calls for completing and utilizing the International Space Station (ISS), establishing an extended human presence on the moon, and preparing for the Mission to Mars, all of which will require sustained habitation of the space environment. In view of these plans, it is vital that we improve our understanding of acute and enduring effects of space travel on the female reproductive system and that we do so soon. The first manned missions to Mars are expected to begin as early as 2015. Time is of the essence in identifying potential reproductive risks and beginning to develop countermeasures that may be important for female astronauts who undertake long-duration missions. Because there is almost no knowledge base on the effects of the space environment on the female reproductive system, ground-based animal studies are essential prerequisites Two studies are described in this report. The first focuses on gravity's effects on adult female reproduction (Ronca, Rushing, Tou, Wade, & Baer, 2005). Typically, fertility cycles of female astronauts are suppressed, so there have been no studies of human female reproductive processes under spaceflight conditions. However, environmental conditions are known to exert several adverse effects on the ovaries and on the hypothalamic-pituitary axis that regulates the ovulatory cycle and is highly susceptible to environmental factors. Stress alone can lead to ovulatory failure. Female mammals undergo characteristic hormonal and cellular changes during the reproductive cycle. In rats, the estrous cycle averages 4-5 days in length and consists of four identifiable phases: proestrus, estrus, metestrus, and diestrus. Numerous environmental factors can interrupt the estrous cycle, such as stress, sterile mating, mechanical stimulation of the cervix, or continuous illumination. It was previously reported that, when centrifugation was initiated during diestrus, the length of the rat's estrous cycle increased due to prolonged diestrus. Female rats exposed to hindimb unloading to mimic aspects of microgravity exposure showed estrous cycle lengthening due to prolongation of the diestrus phase of the cycle. In the present study, we hypothesized that continuous, long-duration (2 mo) exposure to 2-g centrifugation would alter patterns of estrous cycling and delay conception. The NASA Ames Research Center 24-foot Centrifuge was used to produce 2-g. Sexually mature female and male rats weight-matched then assigned to either 2-g centrifuge or non-centrifuge (stationary 1-g control) conditions, and to non-breeding or breeding conditions. In non-breeding females, estrous cycles were analyzed by examining vaginal cytology before and for 35 days during centrifugation. Females in the breeding condition were time-mated following 7 days of adaptation to centrifugation. Following adaptation to centrifugation, both centrifuged and non-centrifuged breeding females conceived, however centrifuged females took four-times longer than controls to achieve conception. Our results indicate that 2-g centrifugation of female rats delayed conception and diminished pregnancy outcome. Thus, long-duration exposure to altered gravity clearly affected mammalian reproductive function. One important question is whether this delay was due to changes in male reproductive function, female reproductive function, or both. To address this issue, we undertook a collaborative effort with Dr. Joseph Tash from the University of Kansas Medical School to examine sperm motility and testes weight and morphology in the male rats exposed to 2-g centrifugation and mated with the females in our study. These analyses revealed no obvious morphological or functional impairments in the males. To identify potential problems with the rats' copulatory behavior, we videotaped mating pairs in our study throughout the breeding period. One objective that we will be able to achieve with the forthcoming reduced funding (approx $51,000 total costs) will be completion of a behavioral analysis of the breeding sequences to determine whether impairments in the males' mounting behavior may have contributed to the delayed conception by the females. Based on the evidence we have accrued thus far, we believe that delayed conception is a sign of ovarian dysfunction in the 2-g exposed females. It is important to determine whether ovarian dysfunction continues following long duration exposure to altered gravity and whether changes in ovarian function are transmitted across generations. The second major goal of our work is to further our understanding of altered gravity as a chronic stressor in females.

	Stress induced by altered gravity may underlie delayed conception. Previously, our work has incorporated a contemporary perspective in maternal-fetal medicine: 'Intrauterine programming' is a newly emerging field that is revealing astounding insights into the prenatal origins of adult disease arising from environmental factors. Adult disease states that have been implicated include metabolic, endocrine, and cardiovascular pathophysiologies. Our results indicate that 2-g centrifugation beginning prior to conception leads to reduced body weights of neonatal offspring, increased body weights and hormonal alterations in adulthood even when animals are reared at 1-g. Collectively, these findings are helping to establish 2-g centrifugation as a unique model for chronic stress that induces programming effects on progeny reared at 1-g. Mere exposure of the female reproductive system (prior to conception) may be sufficient to induce persistant changes in the health and well-being of offspring. Additional hormonal and neural studies are needed to clarify effects of altered gravity as a chronic stressor, including both acute and persistent effects on the female reproductive system, and to identify meaningful countermeasures for use in female astronauts. These concerns are vital to our future space endeavors. In summary, our findings provide some preliminary, albeit important, analyses of altered gravity effects on reproductive physiology of the female mammal. The Vision for Space Exploration calls for conducting space biology studies on the astronauts themselves in future flights rather than relying on animal models to ascertain the physiological consequences of space travel. However is it vital that we understand in advance of long-term habitation of space by female astronauts the requisite changes in reproductive function, whether these changes persist after return to Earth, and mechanisms of cross-generational transmission.
Bibliography Type:	Description: (Last Updated: 05/15/2025)
Abstracts for Journals and Proceedings	Ronca AE, Bombick D, Conway GC, Baer LA. "Differential patterns of gene expression in newborn rat brain. " American Psychological Society "Hot Topic" Presentation, Los Angeles, CA, May 20-26, 2005. Los Angeles, CA, May 20-26, 2005. , May-2005
Abstracts for Journals and Proceedings	Lintault L, Zakrzewska E, Maple R, Baer L, Ronca A, Wade C, Plaut K. "Effects of hypergravity and food intake on gene expression of enzymes in the liver of periparturient rats. " Annual Meeting of the American Society for Gravitational and Space Biology, Reno, NV, November 2005. Gravit Space Biol. 2005 Nov;19(1):20. , Nov-2005
Abstracts for Journals and Proceedings	Zakrzewska E, Lintault L, Casey T, Maple R, Ronca A, Baer L, Wade C, Plaut K. "Can prolactin supplementation alleviate impaired metabolic activity of lactating rats exposed to hypergravity? " Annual Meeting of the American Society for Gravitational and Space Biology, Reno NV, November 2005. Gravit Space Biol. 2005 Nov;19(1):62. , Nov-2005
Abstracts for Journals and Proceedings	Ronca A, Baer L, Rushing L,Wade C. "Antenatal programming of adult body weight in male rats induced by centrifugation is associated with elevated leptin and altered body composition. " Annual Meeting of the American Society for Gravitational and Space Biology, Reno NV, November 2005. Gravit Space Biol. 2005 Nov;19(1):55. , Nov-2005
Abstracts for Journals and Proceedings	Ronca AE, Baer LA, Conatser L, McCue R, Saldanha K. "Labor inside and out: Temporal relationships between intrauterine pressure and behavioral expressions of labor in parturient rats." Annual Meeting of the International Society for Developmental Psychobiology, November 2005. Developmental Psychobiology, 2005;47(4):446. , Nov-2005
Abstracts for Journals and Proceedings	Baer LA, Piper J, Conatser L, Rushing L, Wade CE, Ronca AE. "Fetal programming of adult overweight in male rats induced by 2-g centrifugation." Annual Meeting of the Society for Developmental Psychobiology, November 2005. Developmental Psychobiology. 2005;47(4):417. , Nov-2005
Abstracts for Journals and Proceedings	Ronca AE, Bombick D, Conway GC, Baer LA. "Effects of premature birth on c-fos expression in neonatal rats. " Annual Meeting of the Society for Neuroscience, November 2005. Society for Neuroscience, 35:Online Abstract No. 835.16, November 2005. , Nov-2005
Abstracts for Journals and Proceedings	Patel OV, Lintault L, Zakrzewska E, Maple RL, Baer L, Ronca A, Wade C, Plaut K. "Hypergravity induced suppression of lipid synthesis in periparturient rats: Differential dynamics of prolactin receptor isoform expression following prolactin and bromocryptine supplementation in rats. " Annual Meeting of the Society for the Study of Reproduction, July 2006. Society for the Study of Reproduction, July 2006. , Jul-2006
Articles in Peer-reviewed Journals	Ronca AE, Rushing LS, Tou J, Wade CE, Baer LA. "Centrifugation effects on estrous cycling, mating success and pregnancy outcome in rats." J Gravit Physiol. 2005 Jul;12(1):P183-4. , Jul-2005
Articles in Peer-reviewed Journals	Baer LA, Rushing L, Wade CE, Ronca AE. "Prenatal centrifugation: a model for developmental programming of adult body weight? " J Gravit Physiol. 2005 Jul;12(1):P181-2. , Jul-2005
Articles in Peer-reviewed Journals	Alberts JR, Ronca AE. "Development as adaptation: a paradigm for gravitational and space biology." Adv Space Biol Med. 2005;10:175-207. PMID: 16101108 (Vol. 10, Experimentation with Animal Models in Space) , Dec-2005
Books/Book Chapters	Ronca AE. "Studying the mammalian life cycle in space." in "Sourcebook of Model Organisms in the Biomedical Sciences." Ed. M. Conn. New Jersey: Humana Press, in press 2006., Dec-2006
Significant Media Coverage	Ronca AE, Baer L. "Science at the Edge: Segment 4: Breaking Nature's Rules." Space biology segment being aired on The Science Discovery Channel from 2005 – present., Jan-2005